Mission Project Timeline

TestSat-1

Organization:	Test Org
Satellite Class:	Small Satellite
Spacecraft Mass:	200.0 kg
Target Launch:	August 2028
Document Date:	2026-05-02

1. Executive Summary

This document presents a comprehensive project timeline for the **TestSat-1** mission, a Small Satellite with a target launch date of August 2028. The critical path analysis estimates **54 months** from project initiation to launch readiness (with parallel activities), compared to **27 months** available. This indicates a schedule shortfall of **27 months** under nominal assumptions. The overall probability of on-time completion is assessed at **53%** (moderate outlook). 10 compatible launch vehicles were identified.

Metric	Value
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Serial (No Parallel) Duration	66 months
Time Available to Launch	27 months
Schedule Margin	-27 months
On-Time Probability	53%
Compatible Launch Vehicles	10

Table 1 — Timeline Key Metrics

2. Project Timeline (Gantt Chart)

The following chart shows all project phases from concept through launch. Solid bars represent nominal durations; dashed extensions show pessimistic estimates. The red dashed line marks the target launch date.

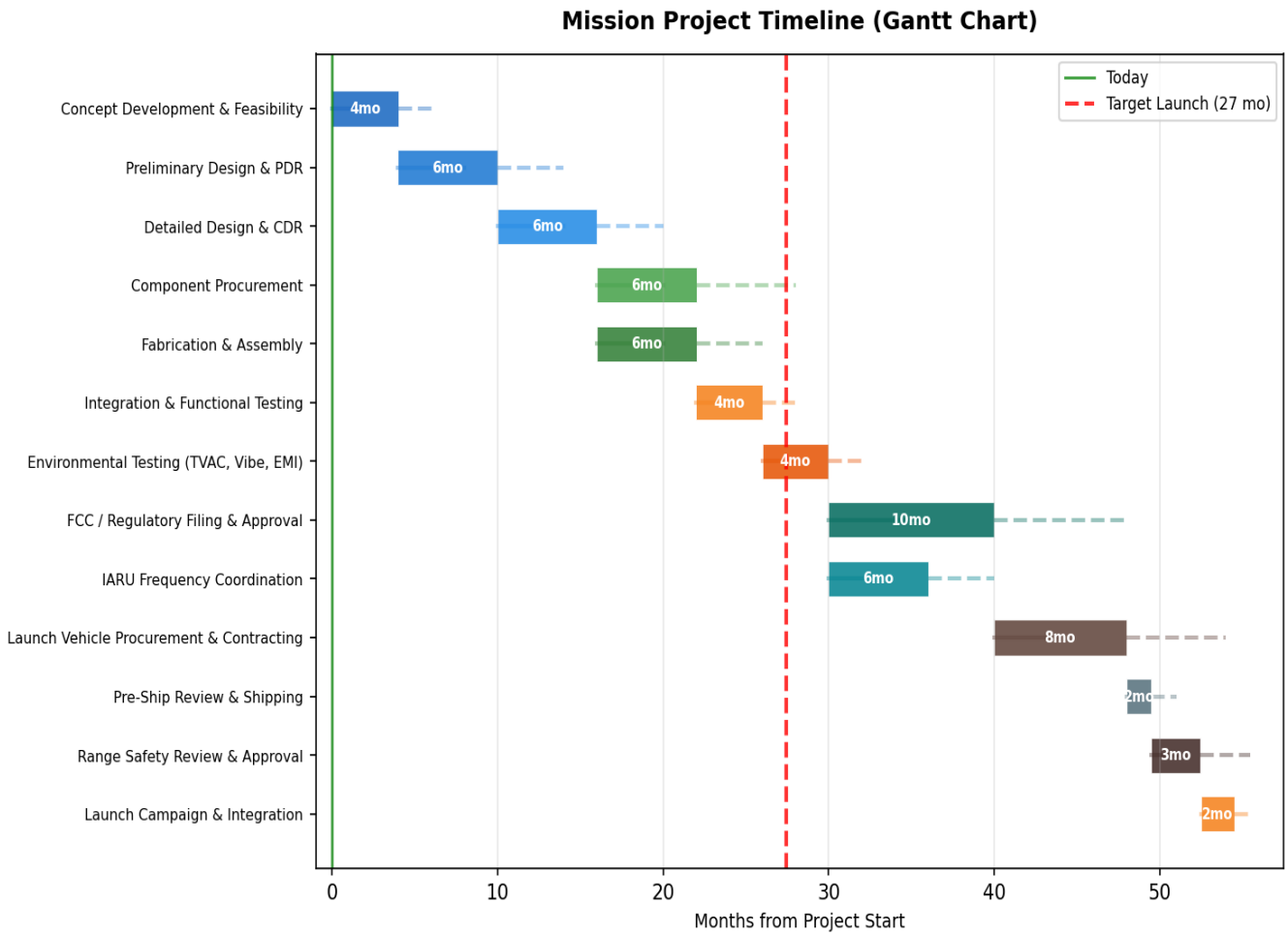

Figure 1 — Mission Project Gantt Chart

3. Phase Details

Each phase includes optimistic, nominal, and pessimistic duration estimates based on industry data for the satellite class and mission parameters.

Phase	Start	End	Opt	Nom	Pess
Concept Development & Feasibility	May 2026	Sep 2026	2.0 mo	4.0 mo	6.0 mo
Preliminary Design & PDR	Sep 2026	Mar 2027	4.0 mo	6.0 mo	10.0 mo
Detailed Design & CDR	Mar 2027	Sep 2027	4.0 mo	6.0 mo	10.0 mo
Component Procurement	Sep 2027	Mar 2028	4.0 mo	6.0 mo	12.0 mo
Fabrication & Assembly	Sep 2027	Mar 2028	4.0 mo	6.0 mo	10.0 mo
Integration & Functional Testing	Mar 2028	Jul 2028	2.0 mo	4.0 mo	6.0 mo
Environmental Testing (TVAC, Vibe, EMI)	Jul 2028	Oct 2028	2.0 mo	4.0 mo	6.0 mo
FCC / Regulatory Filing & Approval	Oct 2028	Sep 2029	6.0 mo	10.0 mo	18.0 mo
IARU Frequency Coordination	Oct 2028	May 2029	3.0 mo	6.0 mo	10.0 mo
Launch Vehicle Procurement & Contracting	Sep 2029	May 2030	4.0 mo	8.0 mo	14.0 mo
Pre-Ship Review & Shipping	May 2030	Jun 2030	1.0 mo	1.5 mo	3.0 mo
Range Safety Review & Approval	Jun 2030	Sep 2030	2.0 mo	3.0 mo	6.0 mo
Launch Campaign & Integration	Sep 2030	Nov 2030	1.0 mo	2.0 mo	3.0 mo

Table 2 — Phase Schedule Detail

3.1 Phase Descriptions

Concept Development & Feasibility (May 2026 — Sep 2026)
Define mission objectives, perform trade studies, establish preliminary requirements, assess technical feasibility, identify key risks, and produce the Mission Concept Review (MCR) package.

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Manufacture custom structural components, assemble subsystems, integrate harnesses, populate circuit boards, and build the flight unit. Includes workmanship inspections.

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Thermal vacuum cycling, vibration (sine and random), shock, EMI/EMC testing per launch vehicle ICD requirements. Includes pre- and post-environmental functional tests.

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Launch Campaign & Integration (Sep 2030 — Nov 2030)
On-site activities at launch facility: spacecraft unpacking, inspection, fueling (if applicable), integration onto launch vehicle, final RF testing, launch rehearsal, and countdown.

4. Launch Vehicle Analysis

Based on the spacecraft mass (200 kg) and target orbit (Sun-Synchronous (SSO)), the following launch vehicles are compatible:

Vehicle	Operator	Capacity	Cost ($M)	Lead Time	Status
ISRO SSLV	ISRO / NSIL	300 kg	$6.0M	6 mo	Operational
Rocket Lab Electron	Rocket Lab	200 kg	$7.5M	12 mo	Operational
SpaceX Transporter (Rideshare)	SpaceX	200 kg	$1.1M	6 mo	Operational
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Table 3 — Compatible Launch Vehicles

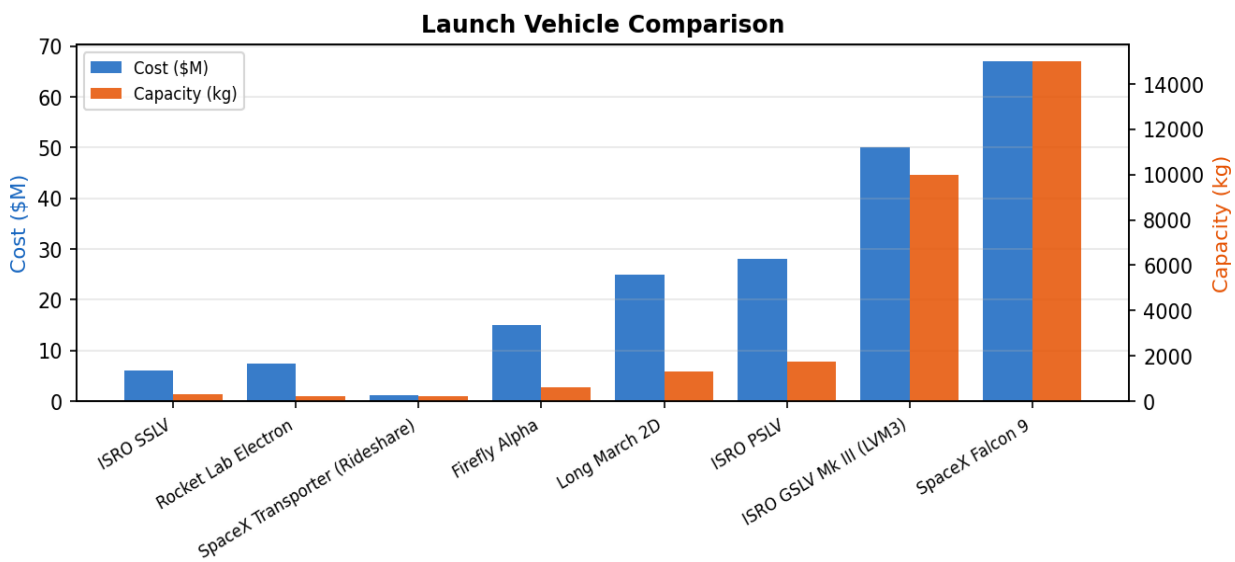

Figure 2 — Launch Vehicle Cost & Capacity Comparison

4.1 Recommended Primary Vehicle

ISRO SSLV by ISRO / NSIL is the top-ranked match based on capacity fit, operational status, and cost efficiency. It offers 300 kg to the target orbit (33% mass margin). Estimated launch cost is $6.0M. Typical procurement lead time is 6 months. Notes: Small satellite launch vehicle. Very low cost, rapid turnaround.

5. Probability of On-Time Completion

The assessed probability of launching on or before August 2028 is **53%** (moderate — active risk management needed).

Figure 3 — On-Time Completion Probability

5.1 Risk Factor Analysis

The probability assessment considers the following factors. Positive values increase confidence; negative values decrease it.

Factor	Impact	Assessment
Schedule Margin	-30	Schedule is 27 months short — major timeline risk
Satellite Complexity	0	Small Satellite — standard complexity
Launch Vehicle Availability	+5	11 operational vehicles available — excellent launch flexibility
Regulatory Path	-2	Standard Part 25.114 — full FCC review process, typically 12+ months

Table 4 — Probability Risk Factors

6. Recommendations

• **Schedule Acceleration Required:** The current timeline exceeds the target launch date by approximately 27 months. Consider: increasing team size, parallel engineering tracks, early long-lead procurement, or adjusting the launch date.

• **Regulatory Filing:** Initiate FCC filing and frequency coordination in parallel with preliminary design. Regulatory approvals are often on the critical path and have unpredictable timelines.

7. Assumptions & Caveats

• Phase durations are based on industry averages for the selected satellite class and may vary significantly based on team experience, funding profile, and supply chain conditions.

• Parallel phase execution assumes adequate staffing and facilities to support concurrent activities.

• FCC processing times are estimates; actual timelines depend on application completeness, FCC backlog, and any requests for additional information.

• Launch vehicle costs and availability are based on publicly available data as of 2025 and are subject to change. Actual pricing requires a formal quote from the launch provider.

• The probability assessment is a heuristic estimate, not a rigorous Monte Carlo analysis. It should be used for planning guidance, not contractual commitments.

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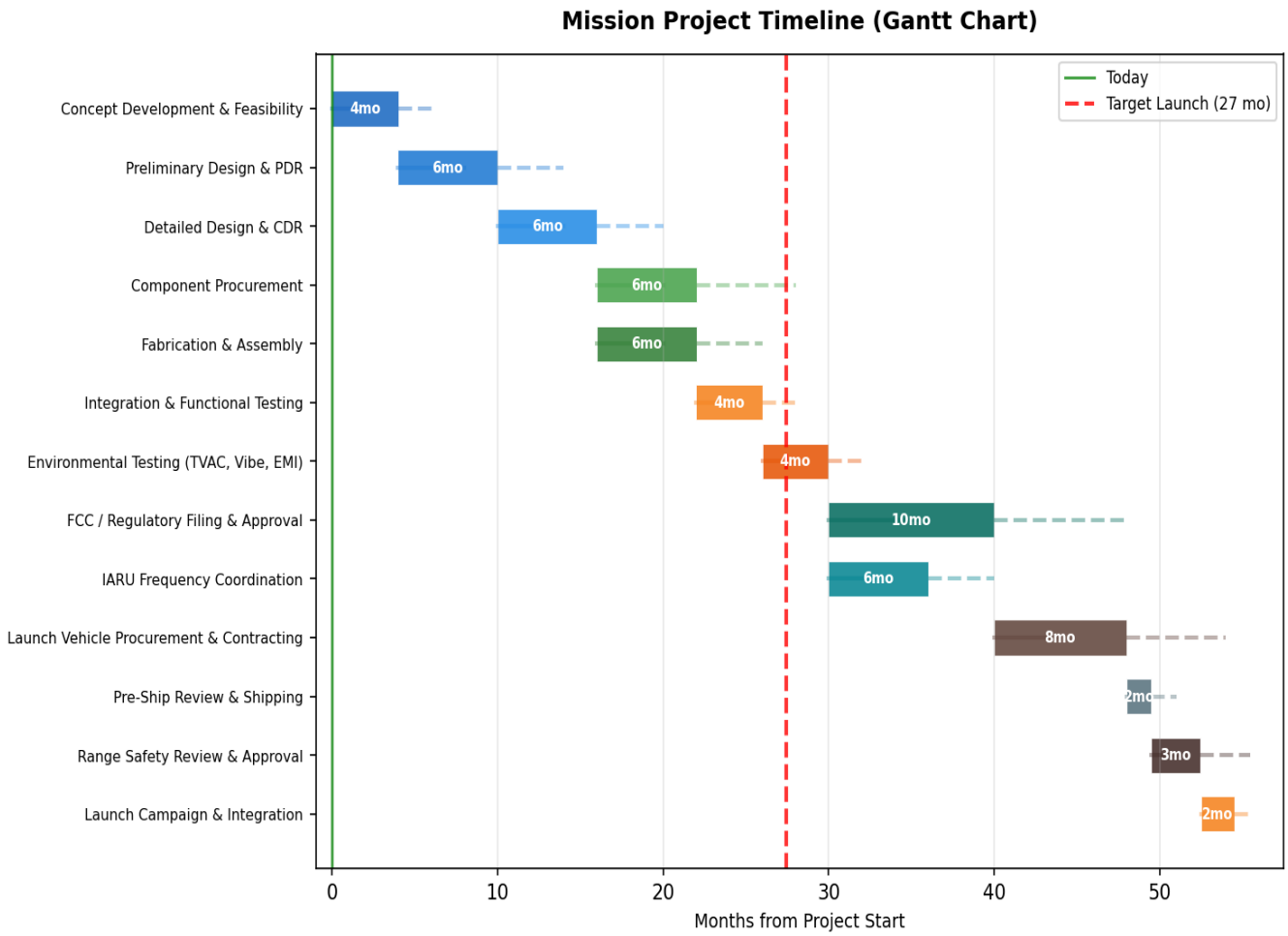

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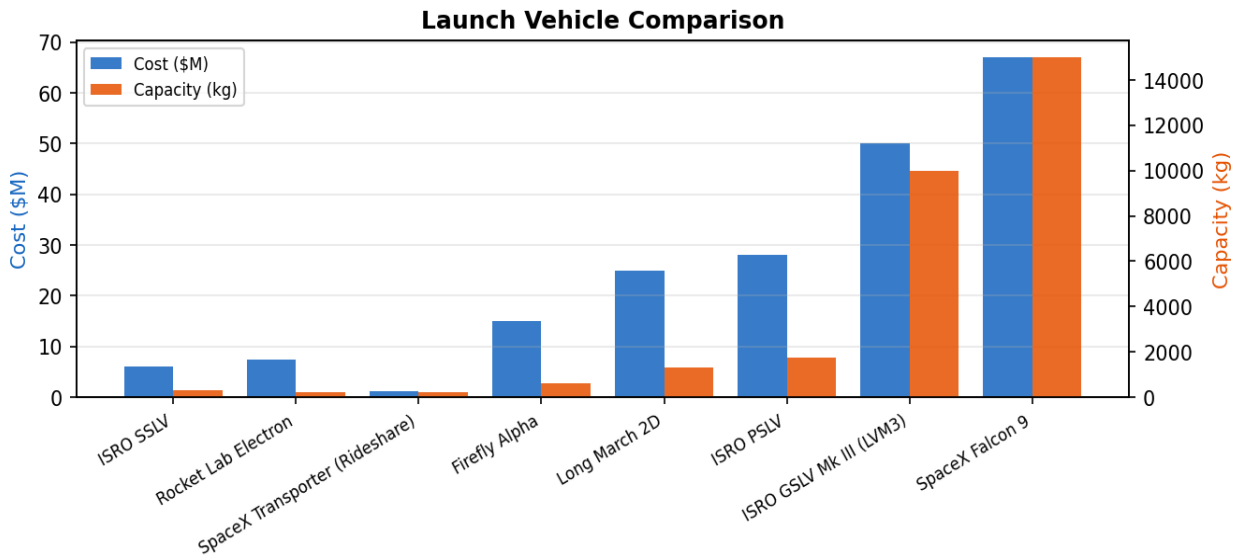

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